

DATA ANALYST

EXCEL PROJECT DOCUMENTATION

RETAIL BUSINESS DATA ANALYSIS USING EXCEL

Project Overview

The Retail Business Analytics Dashboard is designed to analyse sales, customer, product, inventory, and regional performance data. The dashboard provides management with a comprehensive view of business performance through interactive visualizations and key performance indicators (KPIs).

The objective of this project is to transform raw retail data into meaningful insights that support data-driven decision-making.

Business Objective

The main objectives of this project are:

- Monitor overall business performance.
- Analyse customer membership and purchasing behaviour.
- Identify top and low-performing products.
- Track inventory status to prevent stock shortages.
- Evaluate regional sales performance.
- Monitor monthly sales trends.
- Support strategic business decisions using data visualization.

Dataset Description

The project utilizes four datasets:

Customer Data

Customer Data contains customer-related information such as Customer ID & Name, Gender, Membership Level, Age and City.

Product Data

Product Data contains product information such as Product ID, Product Name & Brand, Category, Cost, and Stock Quantity.

Sales Data

Sales data contains sales details such as Sales ID, Customer ID, Product ID, Quantity Sold, Discount applied, Payment Type and Revenue.

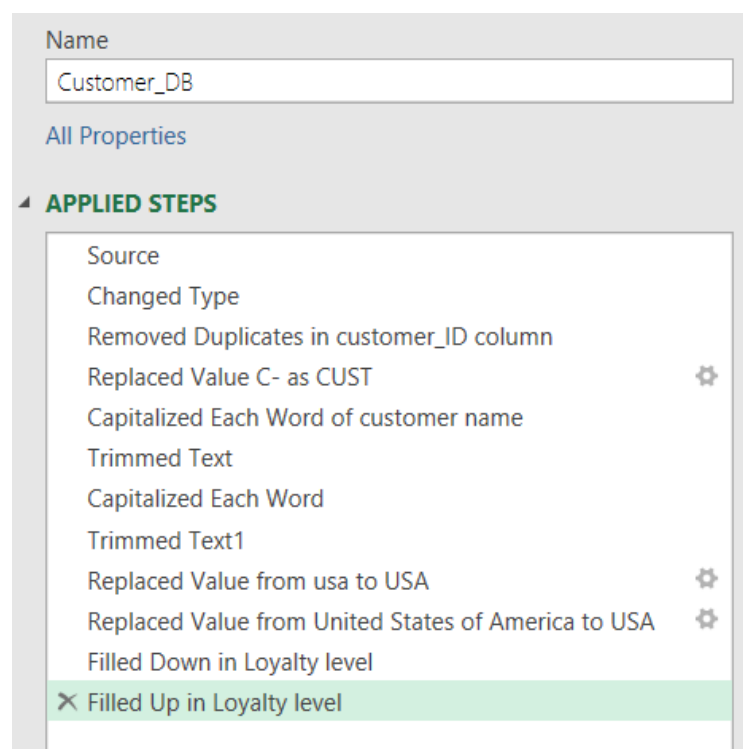
Store Data

Store Data contains store-related information such as Store ID, Region and Store Details.

Data Cleaning with Power Query in Excel:

- Identified and handled missing values by filling blanks with appropriate values.
- Removed duplicate records to maintain data accuracy and consistency.
- Validated the cleaned data to ensure accuracy before creating Pivot Tables and the final dashboard.

1. Customer Database Cleaning Steps:



2. Store Database Cleaning Steps:

Name
Store_DB

[All Properties](#)

APPLIED STEPS

- Source
- Changed Type
- Removed Duplicates
- Trimmed Text
- ✕ Trimmed Text1

3. Product Database Cleaning Steps:

Name
Product_DB

[All Properties](#)

APPLIED STEPS

- Source
- Changed Type
- Removed Duplicates in Product ID column
- Replaced Value P- as PROD ⚙
- Trimmed Text
- Replaced null Value with Median 243 in cost column ⚙
- Replaced null Value with 0 stock column ⚙
- ✕ Changed Type1 as Whole Number

4. Sales Database Cleaning Steps:

Name
Sales_DB

[All Properties](#)

APPLIED STEPS

- Source
- Changed Type
- Removed Duplicates
- Changed Type1-DATE
- Replaced null value as 0 in discount column ⚙
- ✕ Changed Type1

Excel Functions Used

Product Database:

Stock Status:

```
=IF([@Stock]=0,"Out of Stock",  
IF([@Stock]<=267,"Low Stock",  
IF([@Stock]<=400,"Medium Stock","High Stock")))
```

Sales Database Imputation:

1. Quantity Column:

```
=IF(ISBLANK([@Quantity]),([@[Total_Amount]]/[@[Unit_Price]])*(1-[@Discount]),[@Quantity])
```

2. Unit Price Column:

```
=IF(ISBLANK([Unit_Price]),([Total_Amount]/[Quantity])*(1-[Discount]),[Unit_Price])
```

3. Unit Price Column:

```
=IF(ISBLANK([@[Total_Amount]]),[@Column1]*[@Column2]*(1-[@Discount]),[@[Total_Amount]])
```

4. Revenue Status:

```
=IF([@[Total_Amount2]]<=1087,"Low Revenue","High Revenue")
```

Key Performance Indicators (KPIs)

KPI	Value
Total Revenue	2,174,378
Total Transactions	2,000
Total Customers	500
Total Stores	20

KPI	Value
Total Products	100
Total Stock Available	26,529

Pivot Table & Dashboard Components

Membership Distribution

Displays the percentage distribution of customers across:

- Bronze
- Silver
- Gold
- Platinum

Membership vs Revenue

Analyzes revenue contribution from each membership level.

Revenue by Region and Category

Compares category-wise revenue across:

- East
- North
- South
- West

Monthly Sales Trend

Shows monthly revenue performance throughout the year.

Top 5 Products by Revenue

Highlights the highest revenue-generating products.

Bottom 5 Products by Revenue

Identifies products with the lowest revenue contribution.

Inventory Status

Categorizes products into:

- Out of Stock
- Low Stock
- Medium Stock
- High Stock

Revenue by Gender

Compares revenue contribution between male and female customers.

Region-wise Revenue

Displays total revenue generated by each region.

Key Insights

Customer Insights

- Platinum members represent the largest customer segment.
- Platinum members contribute the highest revenue share.
- Female customers generate 57% of total revenue.

Sales Insights

- Total revenue exceeded 2.17 million.
- Sales peaked in March.
- Revenue remained stable during the second half of the year.

Product Insights

- Portable Power Bank generated the highest revenue.
- Beauty products appear among the lowest-performing products.

Inventory Insights

- 47 products are in low stock.
- 3 products are completely out of stock.

- Inventory replenishment is required to avoid stockouts.

Regional Insights

- North region generated the highest revenue.
- West region generated the lowest revenue and requires attention.

Business Recommendations

Inventory Management

- Replenish low-stock and out-of-stock products immediately.
- Implement inventory monitoring systems.

Customer Retention

- Expand loyalty programs to increase Platinum membership.
- Introduce personalized offers for lower-tier members.

Regional Strategy

- Investigate causes of low revenue in the West region.
- Launch targeted marketing campaigns in underperforming regions.

Product Optimization

- Increase promotion of high-performing products.
- Review pricing and marketing strategies for low-performing products.

Conclusion

The Retail Business Analytics Dashboard provides a centralized view of key retail metrics, enabling stakeholders to monitor business performance efficiently. Through customer, product, sales, inventory, and regional analysis, the dashboard helps identify growth opportunities, improve inventory management, and support strategic decision-making.